

Salavat Galimov

Raleigh, NC | (910) 939-0267 | salgalia@proton.me | [trade42.github.io](https://github.com/trade42) | US citizen

Seeking: internship from December 2026 through Summer 2027; open to relocating to San Francisco

Flight videos and brace CAD: <https://drive.google.com/drive/folders/158aF8HUKg1S8hqNXXOVQKNSS-r63nkcE>

EE student who builds embedded hardware end to end: component selection, wiring, firmware, power debugging, and flight testing.

EDUCATION

NC State University, Raleigh, NC | BS Electrical Engineering, second-year (junior standing by credit hours) **Expected May 2029**

Coursework: signals, circuits and systems; computer systems; electricity and magnetism; multivariable calculus.

HARDWARE & ROBOTICS EXPERIENCE

Drop Foot Active Brace | Wearable Robotic Assistive Device (4-person team) **2026 – Present**

- Delivered reliable active foot lift for an adult client with drop foot by building a custom robotic brace: 3 iterations from concept to on-body testing in under 30 days.
- Selected and integrated the electronics on an Arduino Nano (thigh-mounted FSR gait sensor, dual 40 kg-cm servos on a paracord cable drive, 3S Li-ion power); wrote closed-loop C/C++ firmware that calibrates the FSR input and commands the servos; designed the full CAD assembly (3D-printed ankle saddle, motor mounts).
- Ran 6+ debug iterations across power, sensor, and actuator subsystems, including diagnosing and fixing a 5 V rail brownout; evaluated a 12 V DC motor + L298N H-bridge as an alternative actuation path.

Robotics Design Challenge: Lab Automation Cell | Anthrogen (YC S24), Final Interview Round **2026**

- Designed a full lab automation cell in Fusion 360, evolved from an earlier X-rail pipetting gantry: instrument stations, per-station robotic handling, and automated sample transport between stations.

Aerobatic RC Plane | From-Scratch Design, Build, and Flight Test **2021 – 2023**

- Placed 4th at Aerospace Russian Nationals (Camsat Russia, 130+ teams) with a 1.75 m aerobatic RC plane built from scratch (foam airframe, carbon-fiber spars, brushless motors, LiPo); 4 design cycles, 20+ test flights.

Two-Stage Solid-Fuel Rocket with Onboard Telemetry | Aerospace Club **2021 – 2023**

- Placed 2nd at Aerospace Russian Nationals (Camsat Russia, 130+ teams) with a two-stage rocket that reached ~5,000 m apogee; designed in SolidWorks, built, and flown with an onboard IMU, altitude telemetry, and stabilization sensors.

Stealth Startup | AI Topology Optimization for UAV Airframes (Founder) **2026 – Present**

- Building an AI platform that turns rough sketches into weight-optimized UAV airframes and exports them straight to SolidWorks; a GNN surrogate for finite element analysis scores stress on 1,000+ candidates.

SOFTWARE PROJECTS

Vale | Stealth Prop Trading Firm (Founder) **2026 – Present**

- Built a Rust research engine with 8,800+ tests behind a 5-lane CI gate; strategies must clear a statistical validation gate that rejects results explainable by chance, then a sealed out-of-sample holdout.
- Live tick-data pipeline across 25 crypto instruments; every research trial is logged to a hash-chained audit ledger.

Vext | Real-Time Multi-AI Platform (solo build) **2026**

- Async Rust (Axum, Tokio) backend streaming tokens to a Next.js frontend over WebSockets; 47+ Rust tests.

LEADERSHIP & COMPETITIONS

Technology Student Association (TSA) | UAV/Drone Team Lead; VEX Robotics Captain **2023 – 2025**

- Led the Drone/UAV team to 1st at state and 7th nationally: airframe design, build, flight tuning, and mission prep. Captained the chapter's VEX Robotics team to 1st at state.

Aerospace Club | Flight Projects Lead **2021 – 2023**

- Led the UAV team to 3rd at Aerospace Russian Nationals (Camsat Russia, 130+ teams), flying autonomous missions.

TECHNICAL SKILLS & AWARDS

Embedded & Electronics: Arduino (C/C++), Raspberry Pi (Linux); sensor and actuator integration (FSR, IMU, servos, brushless motors, L298N H-bridge); Li-ion and LiPo power systems; wiring and soldering; bench and flight testing; hardware debugging.

Fabrication & CAD: 3D printing, foam and carbon-fiber airframe fabrication; SolidWorks, Fusion 360, OnShape, Blender.

Software: Rust (Axum, Tokio, sqlx), Python (pandas, matplotlib, Flask), TypeScript / React / Next.js, Supabase (Postgres, RLS, Auth), WebSockets, Git, Linux (Arch); AI-assisted engineering with Claude Code, including Fusion 360 through an MCP bridge.

Awards: Game Changers Hackathon 1st place (\$500, team captain); TSA state champion (Drone/UAV and VEX), 7th nationally.